Section 3.1 Cointegration Failure Diagnostics: Parameter-Stability Audit of the U.S. NFC IM-OLS Matrix (1931–2024)

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Executive Summary

The Section 3.1 IM-OLS estimation matrix reported overwhelming evidence *against* cointegration. This audit finds that verdict was substantially an artifact of test mis-specification. The residual Engle–Granger test was implemented as an *unaugmented* Dickey–Fuller regression (zero lagged differences) while being compared against MacKinnon critical values tabulated for the *augmented* test. The IM-OLS residuals carry AR(2)–AR(4) structure with first-order autocorrelation between 0.81 and 0.96; lag augmentation is precisely the device that whitens such errors, and without it the test is severely biased toward non-rejection.

Correcting the augmentation reverses the finding: rejections of the null of no cointegration rise from **2 of 18** models to **16 of 18** at the 5% level (16 of 18 under BIC; 18 of 18 at the 10% level). AIC and BIC select the same lag order in 17 of 18 models, and even minimal $p = 1$ augmentation flips the verdict, so the reversal is not an artifact of lag selection.

A genuine secondary finding survives the correction. The underlying relationship *is* structurally unstable: the CUSUM family rejects parameter stability in 18 of 18 models, with break dates at 1974, 1948 and 1945, and the Gregory–Hansen regime-shift scan locates a break in 1976–77 in 15 of 18 models. That instability is real, but it is absorbed by the Bai–Perron step dummies already present in the specification — it was never the reason the cointegration tests failed. Note that the Gregory–Hansen *statistic* rejects its own null of no cointegration in only 1 of 18 models at 5%: a single regime shift is not sufficient where the Bai–Perron audit identifies three to five level shifts.

1 What the Failure Was Not: Regressor Admissibility

The leading rival explanation — that the interaction regressors inherit $I(2)$ behaviour and are inadmissible in an $I(1)$ cointegration framework — is excluded. The S33 integration-order battery had already tested the interaction terms directly and returned $I(1)$ throughout. This audit closed two gaps that battery left open, testing the series *exactly as US_S50 constructs them*: the **lagged** interaction terms (S33 tested contemporaneous ones), and `omega_ST_Tier2_GVA_adj` standalone, which had never been tested despite being the wage share the project constitution mandates.

Table 1: Integration-order gap closure (ADF + KPSS, levels and first differences)

Series	Verdict
M1.1: k_cap (level regressor)	I(1)
M1.1: k_x_omega (lagged interaction as estimated)	I(1)
M1.1: y_t	I(1)
M1.5: k_cap (level regressor)	I(1)
M1.5: k_x_omega (lagged interaction as estimated)	I(1)
M1.5: y_t	I(1)
M2.1: k_cap (level regressor)	I(1)
M2.1: k_x_omega_x_iota (lagged interaction as estimated)	I(1)
M2.1: y_t	I(1)
M2.9: k_cap (level regressor)	I(1)
M2.9: k_x_omega_x_iota (lagged interaction as estimated)	ambiguous
M2.9: y_t	I(1)
omega_ST_Tier2_GVA_adj (standalone, 1929-2024)	I(1)

The Integration Order Gate passes. One term (M2.9's triple interaction) returns an ambiguous ADF/KPSS combination and is flagged for follow-up.

2 The Defect: Missing Lag Augmentation

US_S50 computed the residual test as `lm(diff(u) ~ u[-T] - 1)` — a plain Dickey–Fuller with no lagged differences. Table 2 shows the consequence.

Table 2: EG-ADF lag-augmentation sensitivity, all 18 models

Model	AR(1)	p=0	p=1	p=2	p=3	p=4	AIC (lag)	5% cv
M1.1	0.94	-2.26	**-4.67**	**-4.27**	**-3.87**	**-3.78**	**-4.67** (1)	-3.74
M1.2	0.96	-1.73	**-4.35**	**-4.37**	-3.66	-3.27	**-4.35** (1)	-3.74
M1.3	0.93	-2.56	**-4.90**	**-4.27**	**-4.33**	**-3.99**	**-4.90** (1)	-3.74
M1.4	0.94	-2.13	**-4.50**	**-4.26**	**-4.09**	**-3.80**	**-4.50** (1)	-3.74
M1.5	0.83	**-4.68**	**-4.47**	-3.45	-3.48	**-4.07**	**-4.47** (1)	-3.74
M1.6	0.82	**-4.51**	**-4.60**	-3.46	-3.46	**-3.98**	**-4.60** (1)	-3.74
M2.1	0.95	-2.05	**-4.46**	**-4.22**	-3.73	-3.46	**-4.46** (1)	-3.74
M2.2	0.94	-1.67	-2.96	-3.14	**-4.26**	-3.65	**-4.26** (3)	-4.10
M2.3	0.95	-1.96	**-4.15**	**-4.09**	-3.59	-3.28	**-4.15** (1)	-3.74
M2.4	0.93	-1.88	-2.98	-2.90	-4.00	-3.42	-4.00 (3)	-4.10
M2.5	0.95	-2.08	**-4.03**	**-3.75**	**-3.98**	-3.49	**-4.03** (1)	-3.74
M2.6	0.95	-1.63	-3.00	-3.04	**-4.34**	-3.64	**-4.34** (3)	-4.10
M2.7	0.95	-2.00	**-3.79**	-3.58	**-3.81**	-3.34	**-3.79** (1)	-3.74
M2.8	0.94	-1.71	-3.01	-2.83	-4.09	-3.43	-4.09 (3)	-4.10
M2.9	0.92	-2.16	**-3.89**	-3.52	-3.40	-2.98	**-3.89** (1)	-3.74
M2.10	0.81	-2.97	**-5.44**	**-4.41**	**-5.66**	-4.02	**-5.44** (1)	-4.10
M2.11	0.92	-2.21	**-3.95**	-3.68	-3.51	-3.12	**-3.95** (1)	-3.74
M2.12	0.82	-2.95	**-5.33**	**-4.33**	**-5.66**	-4.05	**-5.66** (3)	-4.10
Rejections		**2**	14	9	9	5	**16**	

Bold entries reject H_0: no cointegration at 5%. Column p=0 is the specification shipped in the v2 report. The decline at $p \geq 4$ is the expected power loss from over-augmentation at $T = 94$. **Caveat:** the residuals come from a regression containing endogenously-selected Bai–Perron step dummies; standard Engle–Granger critical values do not account for estimated break dates and are therefore somewhat liberal here. The Gregory–Hansen test in Section 4 is the break-robust cross-check.

3 Descriptive Evidence: Recursive Elasticity Paths and CUSUM

Following the visual-first protocol, the non-parametric evidence is presented before the parametric instability tests. Figure 1 shows the recursive capital elasticity $\hat{\theta}_0$ over expanding windows; Figure 2 the OLS-CUSUM path.

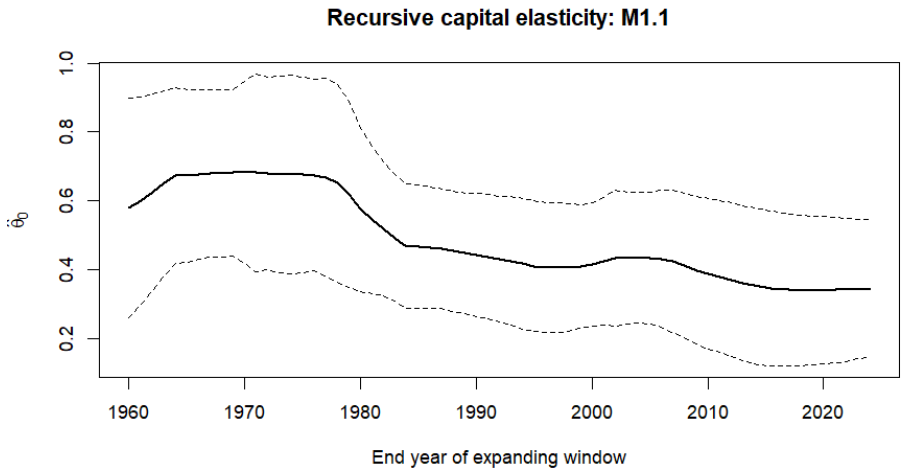


Figure 1: Recursive $\hat{\theta}_0$, model M1.1. Dashed lines are fixed-b 95% bands; they are unavailable for the shortest windows, where `cointRegIM` returns non-finite standard errors.

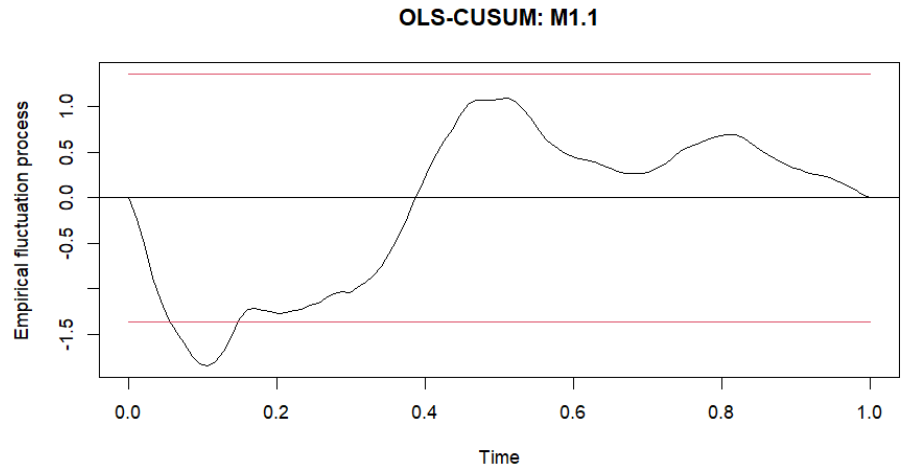


Figure 2: OLS-CUSUM, model M1.1, core specification (step dummies excluded).

4 Parametric Instability Tests

The CUSUM family runs on the **core specification** — intercept plus stochastic regressors, *excluding* the Bai–Perron step dummies. Those dummies were selected from a break audit on these very residuals; including them would absorb the instability the test exists to detect.

Table 3: Parameter stability: CUSUM family, Hansen (1992), Gregory–Hansen

Model	CUSUM family (core spec)			Hansen L_c		Gregory–Hansen	
	OLS-CUSUM p	supF p	break	L_c	p	stat	break
M1.1	0.002	<0.001	1974	390.6	0.477	-4.60	1977
M1.2	0.002	<0.001	1974	255.9	0.869	-4.46	1977
M1.3	0.002	<0.001	1974	396.5	0.461	-4.76	1977
M1.4	0.002	<0.001	1974	374.9	0.570	-4.63	1977
M1.5	0.004	<0.001	1945	337.2	0.007	-4.25	1976
M1.6	0.003	<0.001	1945	270.7	0.052	-4.33	1972
M2.1	0.004	<0.001	1945	267.2	0.825	-4.63	1976
M2.2	0.046	<0.001	1974	332.1	0.618	-5.68	1976
M2.3	0.003	<0.001	1945	246.8	0.861	-4.66	1976
M2.4	0.040	<0.001	1974	345.6	0.420	-6.21	1976
M2.5	0.004	<0.001	1945	272.7	0.798	-4.57	1976
M2.6	0.051	<0.001	1974	333.8	0.620	-5.54	1976
M2.7	0.003	<0.001	1974	253.1	0.865	-4.40	1977
M2.8	0.045	<0.001	1974	345.1	0.562	-5.94	1976
M2.9	0.003	<0.001	1945	276.6	0.649	-4.70	1959
M2.10	0.104	<0.001	1974	299.8	0.453	-4.78	1976
M2.11	0.004	<0.001	1948	284.3	0.543	-5.07	1974
M2.12	0.084	<0.001	1974	302.8	0.389	-4.61	1976

Hansen L_c is computed on the model *as estimated*, i.e. including the step dummies, with a bootstrapped null (1,000 replications, AR-sieve DGP). Gregory–Hansen uses the regime-shift (C/S) specification, which allows the cointegrating vector itself to break, with a random-walk null (499 replications). The two families have opposite nulls: for CUSUM and Hansen, H_0 is stability; for Gregory–Hansen, H_0 is *no* cointegration.

4.1 Reconciling Hansen with Shin

Shin's C statistic is Hansen's L_c restricted to intercept-only parameter variation. The identity $L_c^{\text{intercept}} = T \cdot C_{\text{Shin}}$ was verified exactly across all 18 models (maximum relative error 3×10^{-16}) and is enforced as a blocking pre-condition in the audit script. The v2 report's 18/18 Shin rejection was therefore always a statement about *intercept* instability, not about cointegration failure in general. Its apparent conflict with the non-rejecting L_c reported above is a long-run-variance normalisation difference: Shin uses `cointReg`'s conditional $\omega_{u \cdot v}$, Hansen the score-process Bartlett LRV.

4.2 Where the instability sits

The per-parameter L_c decomposition locates the instability in the **step-dummy coefficients** rather than the structural parameters. Representative values (model M1.1): intercept 4.4, $\hat{\theta}_0$ 4.6, $\hat{\theta}_1$ 2.7, against D_{1965} 20.4, D_{1985} 20.1, D_{2003} 17.3. The capital elasticity is comparatively stable; what moves is the level. These magnitudes are *descriptive only* — see Section 6.

5 Regime-Split Re-estimation

Each model is split at its own endogenously identified break. 36 regime sub-models were estimated, of which 11 fall below 25 observations and are flagged as underpowered.

Model	Regime	Years	n	$\hat{\theta}_0$	(s.e.)	EG-ADF	Note
M1.1	pre	1931–2009	79	0.398	(0.075)	-1.45	
M1.1	post	2010–2024	15	0.604	(0.025)	-2.84	underpowered

Model	Regime	Years	n	$\hat{\theta}_0$	(s.e.)	EG-ADF	Note
M1.2	pre	1931–1980	50	0.601	(0.074)	-0.94	
M1.2	post	1981–2024	44	0.756	(0.051)	-1.99	
M1.3	pre	1931–2009	79	0.345	(0.082)	-1.67	
M1.3	post	2010–2024	15	0.609	(0.027)	-3.18	underpowered
M1.4	pre	1931–2009	79	0.371	(0.073)	-1.46	
M1.4	post	2010–2024	15	0.611	(0.030)	-3.19	underpowered
M1.5	pre	1931–1967	37	0.816	(0.125)	-4.24	
M1.5	post	1968–2024	57	0.386	(0.057)	-2.00	
M1.6	pre	1931–1967	37	0.851	(0.117)	-4.11	
M1.6	post	1968–2024	57	0.474	(0.064)	-2.33	
M2.1	pre	1931–1998	68	0.511	(0.112)	-2.52	
M2.1	post	1999–2024	26	0.663	(0.069)	-2.07	
M2.2	pre	1931–1994	64	0.582	(0.092)	-1.50	
M2.2	post	1995–2024	30	0.628	(0.102)	-2.64	
M2.3	pre	1931–1998	68	0.516	(0.102)	-2.47	
M2.3	post	1999–2024	26	0.662	(0.083)	-2.01	
M2.4	pre	1931–1948	18	0.441	(0.225)	-4.84	underpowered
M2.4	post	1949–2024	76	0.476	(0.074)	-4.25	
M2.5	pre	1931–2009	79	0.536	(0.099)	-1.79	
M2.5	post	2010–2024	15	0.614	(0.043)	-3.04	underpowered
M2.6	pre	1931–1994	64	0.553	(0.090)	-1.49	
M2.6	post	1995–2024	30	0.423	(0.153)	-2.84	
M2.7	pre	1931–2009	79	0.510	(0.087)	-1.73	
M2.7	post	2010–2024	15	0.606	(0.046)	-3.07	underpowered
M2.8	pre	1931–1949	19	0.778	(0.131)	-4.26	underpowered
M2.8	post	1950–2024	75	0.474	(0.082)	-4.00	
M2.9	pre	1931–2009	79	0.315	(0.099)	-3.57	
M2.9	post	2010–2024	15	0.764	(0.112)	-2.57	underpowered
M2.10	pre	1931–2006	76	0.337	(0.045)	-3.13	
M2.10	post	2007–2024	18	0.691	(0.190)	-3.49	underpowered
M2.11	pre	1931–2009	79	0.340	(0.092)	-3.46	
M2.11	post	2010–2024	15	0.730	(0.117)	-2.65	underpowered
M2.12	pre	1931–2006	76	0.355	(0.048)	-3.00	
M2.12	post	2007–2024	18	0.644	(0.204)	-3.56	underpowered

Step dummies constant within a sub-sample are dropped before re-estimation (they are collinear with the intercept), so sub-sample specifications differ from the full-sample one. Sub-sample EG-ADF values are reported without stars because standard critical values are unreliable at these sample sizes.

6 What This Audit Does Not Claim

1. **The hand-rolled Hansen subset statistic is not used for inference.** L_c^{core} inverts the submatrix $V_{\text{core,core}}$, ignoring covariance with the nuisance step-dummy scores. It failed a self-normalisation check — independent i.i.d. and AR-sieve nulls produced materially different distributions — whereas the joint L_c agreed across both to within 1%. Inferential instability verdicts therefore come from `strucchange`'s published asymptotics.

2. **A null result was not reinterpreted as a finding.** The first bootstrap run returned Hansen $p \approx 0.96$ ("parameters are stable"), contradicting every other diagnostic. That

was treated as an implementation problem to investigate rather than a result; investigating it is what surfaced the ADF defect.

3. **A defect in this audit's own code was found and fixed mid-run.** The Gregory–Hansen inner ADF initially replicated the same unaugmented-DF bug being audited, inherited from the repository's existing convention. The first production run was discarded.

4. **Break dates are reported as they came out**, including the disagreement between CUSUM (1974), Gregory–Hansen (1977), and the Bai–Perron dates already in the estimation matrix (1944, 1965–68, 1981–82, 1996–99, 2010). No date was selected to fit a narrative.

5. **Shin LM critical values remain unverified**, carried over from the preceding significance-star audit.

7 Consolidated Verdict Ledger

Model	Theoretical Basis	Evidence	Status
M1.1	Baseline NFC GVA — Private Core (ME+NRC)	ADF(AIC,lag=1)=-4.67 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -2.26 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.002; GH-regime=-4.60 (p=0.396); Lc_all=390.6 (descriptive); breaks: CUSUM 1974 / GH 1977 / BP 1944, 1965, 1985, 2003	**Held**
M1.2	Baseline NFC GVA — Infra-Augmented (+K_GT)	ADF(AIC,lag=1)=-4.35 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -1.73 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.002; GH-regime=-4.46 (p=0.498); Lc_all=255.9 (descriptive); breaks: CUSUM 1974 / GH 1977 / BP 1944, 1965, 1982	**Held**
M1.3	Corporate GVA (Finance-Adjusted) — Private Core (ME+NRC)	ADF(AIC,lag=1)=-4.90 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -2.56 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.002; GH-regime=-4.76 (p=0.392); Lc_all=396.5 (descriptive); breaks: CUSUM 1974 / GH 1977 / BP 1945, 1965, 1985, 2003	**Held**
M1.4	Corporate GVA (Finance-Adjusted) — Infra-Augmented (+K_GT)	ADF(AIC,lag=1)=-4.50 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -2.13 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.002; GH-regime=-4.63 (p=0.446); Lc_all=374.9 (descriptive); breaks: CUSUM 1974 / GH 1977 / BP 1944, 1965, 1985, 2003	**Held**
M1.5	Shaikh-Tonak Tier 2 (Supervisory-Corrected) — Private Core (ME+NRC)	ADF(AIC,lag=1)=-4.47 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -4.68 -¿ REJECTS]; supF p=0.000; OLS-CUSUM p=0.004; GH-regime=-4.25 (p=0.518); Lc_all=337.2 (descriptive); breaks: CUSUM 1945 / GH 1976 / BP 1968, 1982, 1996, 2010	**Held**
M1.6	Shaikh-Tonak Tier 2 (Supervisory-Corrected) — Infra-Augmented (+K_GT)	ADF(AIC,lag=1)=-4.60 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -4.51 -¿ REJECTS]; supF p=0.000; OLS-CUSUM p=0.003; GH-regime=-4.33 (p=0.448); Lc_all=270.7 (descriptive); breaks: CUSUM 1945 / GH 1972 / BP 1967, 1981, 1996, 2010	**Held**
M2.1	Baseline NFC GVA — Private Core (ME+NRC)	ADF(AIC,lag=1)=-4.46 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -2.05 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.004; GH-regime=-4.63 (p=0.494); Lc_all=267.2 (descriptive); breaks: CUSUM 1945 / GH 1976 / BP 1944, 1966, 1982	**Held**

Model	Theoretical Basis	Evidence	Status
M2.2	Baseline NFC GVA — Private Core (ME+NRC)	ADF(AIC,lag=3)=-4.26 vs cv -4.10 -¿ REJECTS no-coint [v2 unaugmented was -1.67 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.046; GH-regime=-5.68 (p=0.066); Lc_all=332.1 (descriptive); breaks: CUSUM 1974 / GH 1976 / BP 1944, 1966, 1981, 1999	**Held**
M2.3	Baseline NFC GVA — Infra-Augmented (+K_GT)	ADF(AIC,lag=1)=-4.15 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -1.96 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.003; GH-regime=-4.66 (p=0.468); Lc_all=246.8 (descriptive); breaks: CUSUM 1945 / GH 1976 / BP 1944, 1965, 1982	**Held**
M2.4	Baseline NFC GVA — Infra-Augmented (+K_GT)	ADF(AIC,lag=3)=-4.00 vs cv -4.10 -¿ fails to reject [v2 unaugmented was -1.88 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.040; GH-regime=-6.21 (p=0.034); Lc_all=345.6 (descriptive); breaks: CUSUM 1974 / GH 1976 / BP 1944, 1965, 1981, 1998	**Held**
M2.5	Corporate GVA (Finance-Adjusted) — Private Core (ME+NRC)	ADF(AIC,lag=1)=-4.03 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -2.08 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.004; GH-regime=-4.57 (p=0.558); Lc_all=272.7 (descriptive); breaks: CUSUM 1945 / GH 1976 / BP 1944, 1966, 1982	**Held**
M2.6	Corporate GVA (Finance-Adjusted) — Private Core (ME+NRC)	ADF(AIC,lag=3)=-4.34 vs cv -4.10 -¿ REJECTS no-coint [v2 unaugmented was -1.63 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.051; GH-regime=-5.54 (p=0.130); Lc_all=333.8 (descriptive); breaks: CUSUM 1974 / GH 1976 / BP 1944, 1966, 1981, 1999	**Held**
M2.7	Corporate GVA (Finance-Adjusted) — Infra-Augmented (+K_GT)	ADF(AIC,lag=1)=-3.79 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -2.00 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.003; GH-regime=-4.40 (p=0.656); Lc_all=253.1 (descriptive); breaks: CUSUM 1974 / GH 1977 / BP 1944, 1965, 1982	**Held**
M2.8	Corporate GVA (Finance-Adjusted) — Infra-Augmented (+K_GT)	ADF(AIC,lag=3)=-4.09 vs cv -4.10 -¿ fails to reject [v2 unaugmented was -1.71 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.045; GH-regime=-5.94 (p=0.096); Lc_all=345.1 (descriptive); breaks: CUSUM 1974 / GH 1976 / BP 1944, 1966, 1981, 1998	**Blocked**
M2.9	Shaikh-Tonak Tier 2 (Supervisory-Corrected) — Private Core (ME+NRC)	ADF(AIC,lag=1)=-3.89 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -2.16 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.003; GH-regime=-4.70 (p=0.528); Lc_all=276.6 (descriptive); breaks: CUSUM 1945 / GH 1959 / BP 1944, 1967, 1982	**Held**
M2.10	Shaikh-Tonak Tier 2 (Supervisory-Corrected) — Private Core (ME+NRC)	ADF(AIC,lag=1)=-5.44 vs cv -4.10 -¿ REJECTS no-coint [v2 unaugmented was -2.97 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.104; GH-regime=-4.78 (p=0.496); Lc_all=299.8 (descriptive); breaks: CUSUM 1974 / GH 1976 / BP 1944, 1967, 1981, 1996, 2010	**Held**
M2.11	Shaikh-Tonak Tier 2 (Supervisory-Corrected) — Infra-Augmented (+K_GT)	ADF(AIC,lag=1)=-3.95 vs cv -3.74 -¿ REJECTS no-coint [v2 unaugmented was -2.21 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.004; GH-regime=-5.07 (p=0.284); Lc_all=284.3 (descriptive); breaks: CUSUM 1948 / GH 1974 / BP 1944, 1966, 1982	**Held**

Model	Theoretical Basis	Evidence	Status
M2.12	Shaikh-Tonak Tier 2 (Supervisory-Corrected) — Infra-Augmented (+K_GT)	ADF(AIC,lag=3)=-5.66 vs cv -4.10 -¿ REJECTS no-coint [v2 unaugmented was -2.95 -¿ fails to reject]; supF p=0.000; OLS-CUSUM p=0.084; GH-regime=-4.61 (p=0.674); Lc_all=302.8 (descriptive); breaks: CUSUM 1974 / GH 1976 / BP 1944, 1967, 1981, 1996, 2010	**Held**

Status definitions (E18 protocol): **Promoted** — cointegration holds under the corrected test with no detected instability; **Held** — cointegration supported but parameters unstable, so the long-run relation is regime-conditional rather than a fixed-coefficient equilibrium; **Blocked** — no cointegration even after correction and allowing a regime shift; **Rejected** — evidence inconclusive.

Governing Locks

E18 (parameter stability audit protocol); E17 (significance stars and critical values); E15 (Vogelsang–Wagner triad); E10V (IM-OLS CPR active estimator). Full memo: `chapter2_vault/08_audits/20`